

GLYPH

Build guide

An off-grid LoRa messenger with GPS and an e-ink screen.



16 steps

No soldering

3 simple tools



Before you start

Tools

-  **Small flat screwdriver** for the screw terminal
-  **Flush cutters** or side cutters
-  **Insulating tape**

Battery safety

The 18650 cell stores a lot of energy. Use an undamaged cell with an intact wrap, keep red with red and black with black, and never let bare wires touch. Put the cell in last.

The build at a glance



What the signs mean



Pay attention: an easy place to go wrong.



Don't force it. Plugs only fit one way.



Check before you go on.



Press until you hear a click.



Move or push this way.



Look here.

A green frame shows how it should look when the step is done.



Parts

ELECTRONICS



1x

LilyGO T3S3 E-Paper



1x

SparkFun GNSS SAM-M8Q



1x

Adafruit LSM6DSOX +
LIS3MDL



1x

Adafruit SHT40



1x

Arduino Modulino Buttons



1x

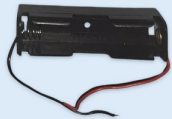
microSD card, 8 or 16 GB

POWER AND CABLES



1x

18650 cell,
3400 mAh



1x

18650 holder



1x

2-pin screw
terminal



1x

Battery cable
(LilyGO plug)



5x

Qwiic cable,
50 mm

ANTENNAS



1x

LoRa antenna



1x

Bluetooth /
Wi-Fi
antenna

PRINTED



1x

Case



1x

Lid



1x

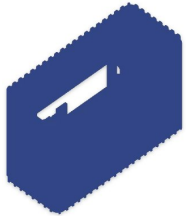
Button cover

Often in the LilyGO box: the antennas and the battery cable. Missing one? Any 2-pin cable for its battery socket, or 2.4 GHz flex antenna with a U.FL plug, fits. Printed: PETG-CF case and lid, TPU button cover.



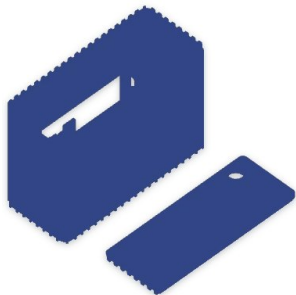
Print the parts

1



Print the **"8.48mm tester print"** first. Check how the LilyGO's gold antenna port fits the hole on top.

Fits snugly

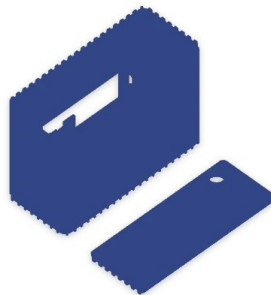


GLYPH - 8.48mm

Too loose



Print the **8.3mm tester print**.
Fits?

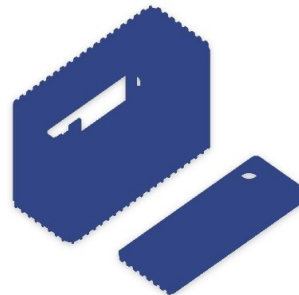


GLYPH-8.3mm

Too tight



Print the **8.66mm tester print**. Fits?



GLYPH - 8.66mm

2



Then print the **button cover** plate in **TPU**.

Already set in the .3mf

Printer	Bambu Lab A1, 0.4 mm nozzle
Profile	0.20 mm Standard
Case and lid	PETG-CF
Button cover	TPU for AMS
Walls · infill	4 · 15%
Supports	On

Each GLYPH plate has the case and the lid. Other printers: keep these settings.

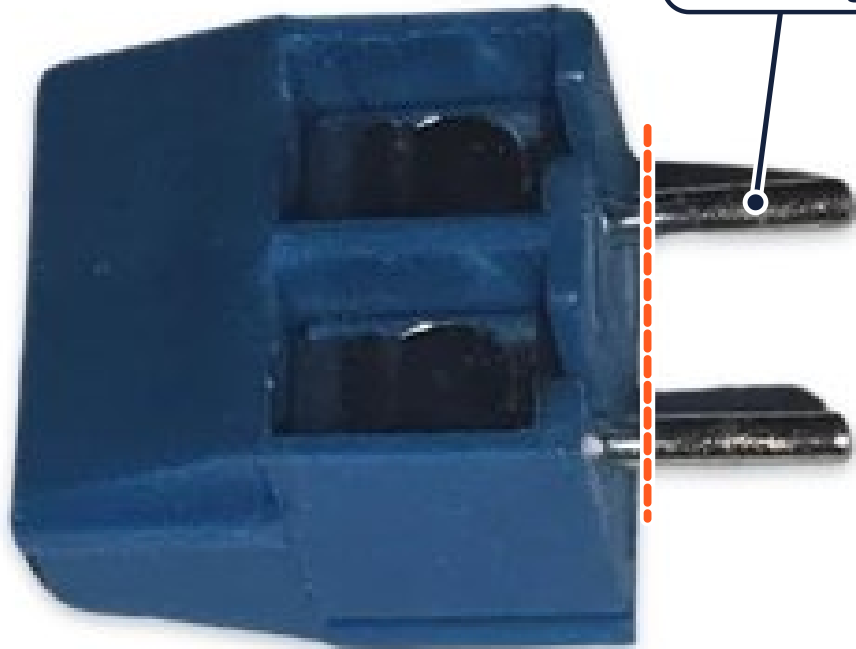
1

Trim the screw terminal



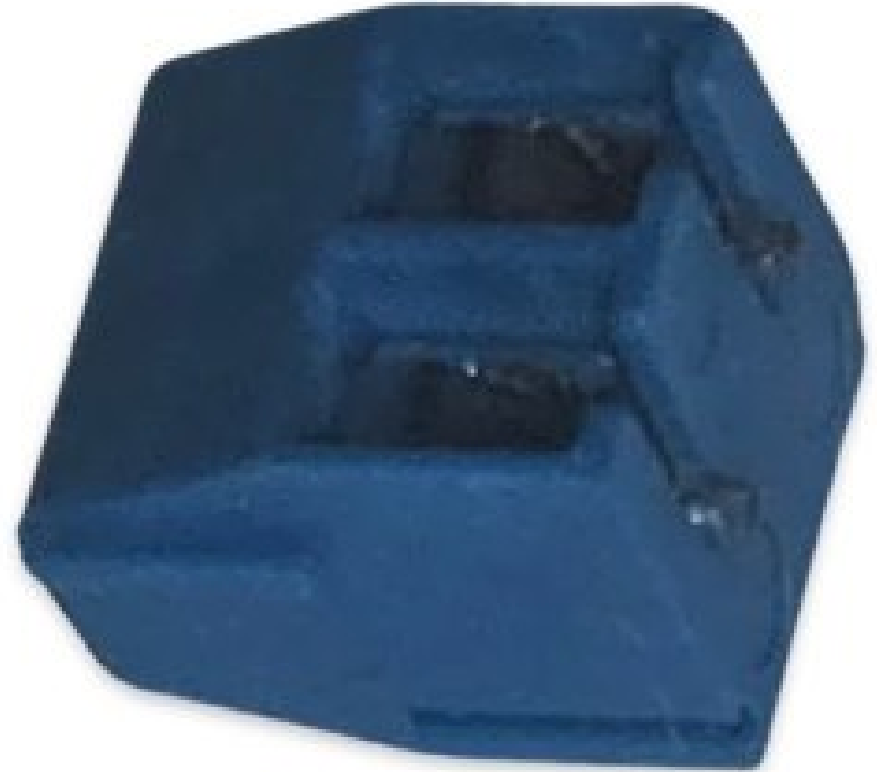
1x

1



Cut off both metal legs, flush with the plastic.

2



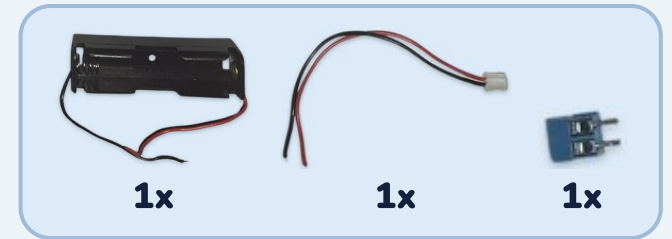
Like this. Nothing sticks out.



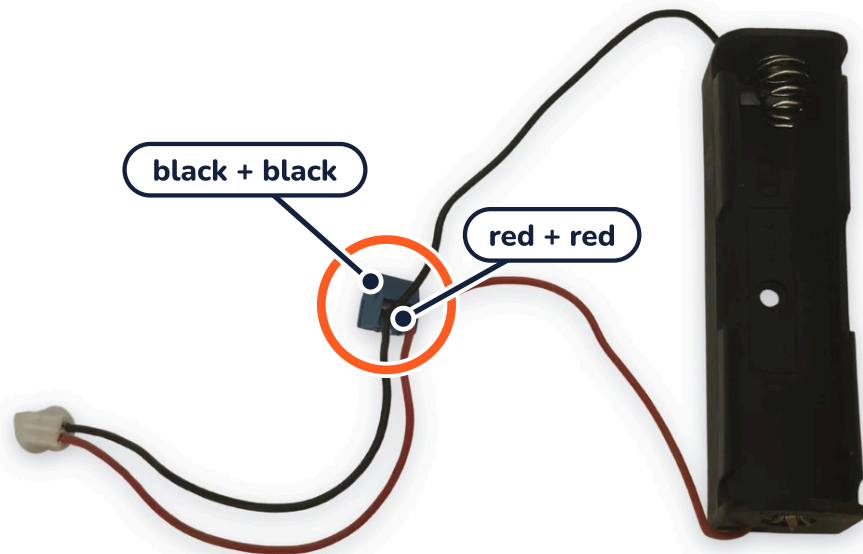
Flush cutters or side cutters.

2

Wire the battery holder

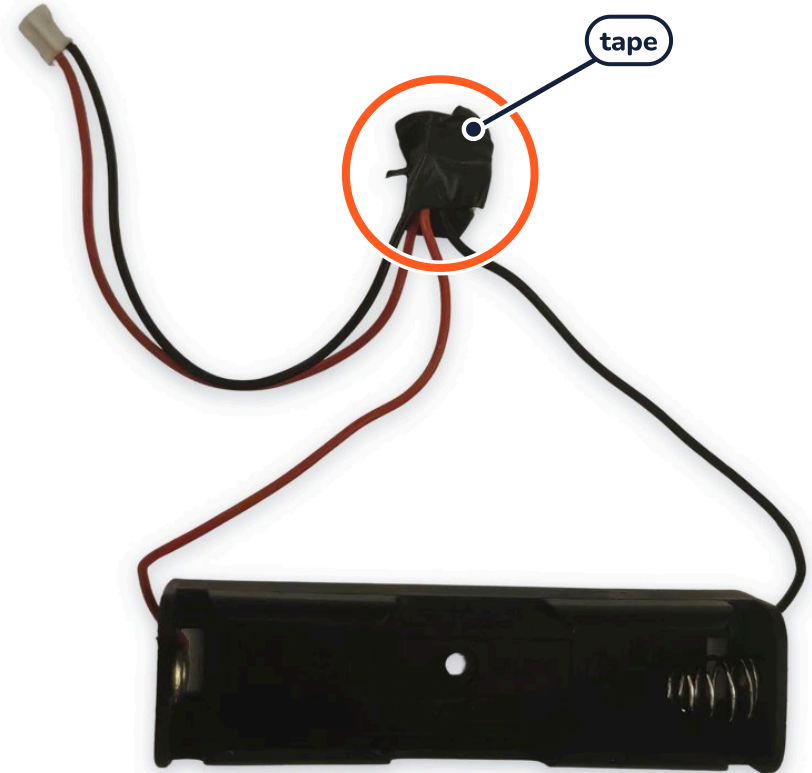


1



Into one side of the terminal: both **black** wires.
Into the other side: both **red** wires. Tighten both screws.

2



Wrap the terminal in insulating tape.

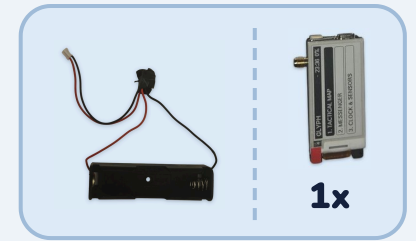


Red with red, black with black. Swapped wires can destroy the board.

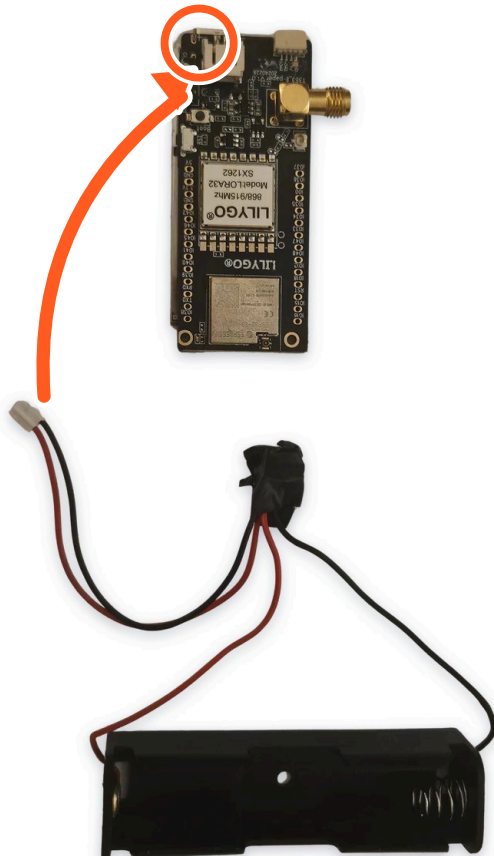


Tug each wire gently: none comes out.

3 Plug the battery into the LilyGO

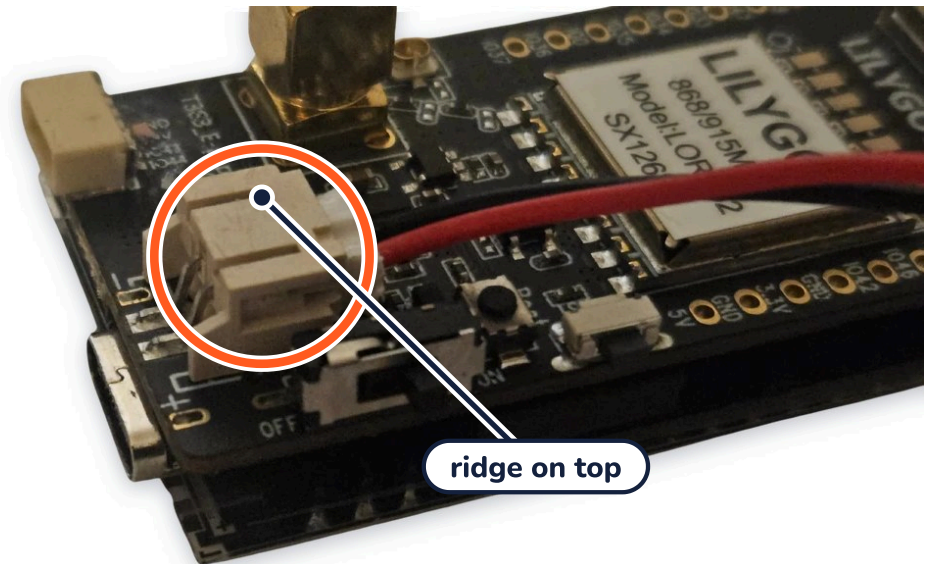


1



Plug the battery cable into this socket.

2

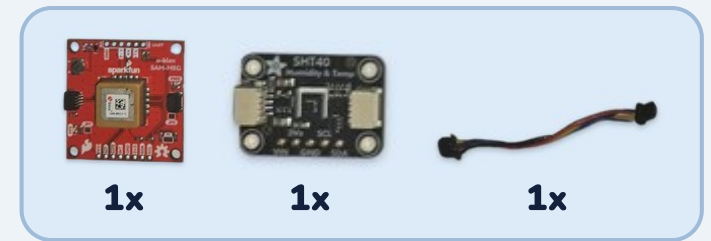


The ridge in the middle of the plug faces up.

✗ Don't force it. If it won't go in, turn the plug over.

4

Join the GPS and the SHT40



1

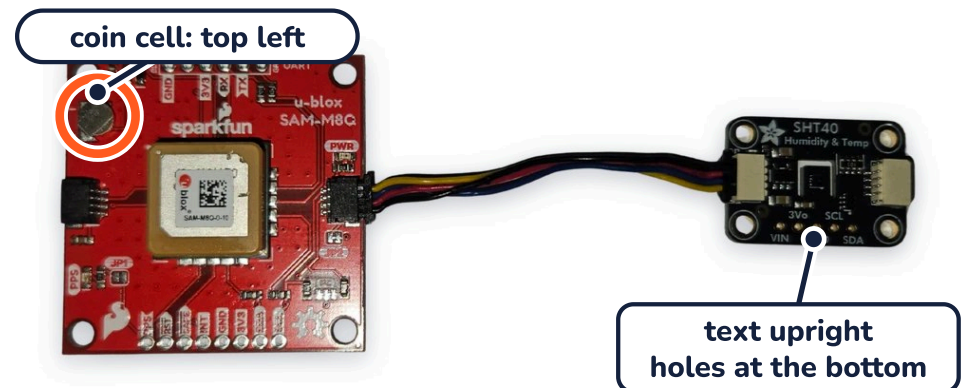


Plug one Qwiic cable between the two boards.



GPS: the small coin cell is **top left**. SHT40: you can read the text, and the VIN · GND · SDA holes are **at the bottom**.

2



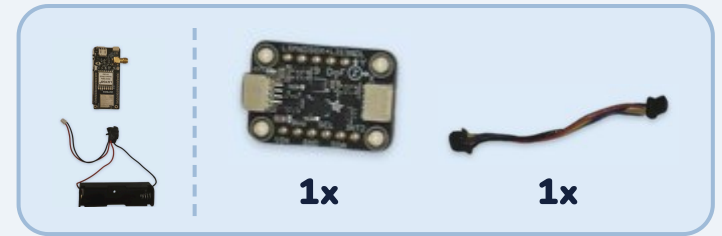
Turn them exactly like this.



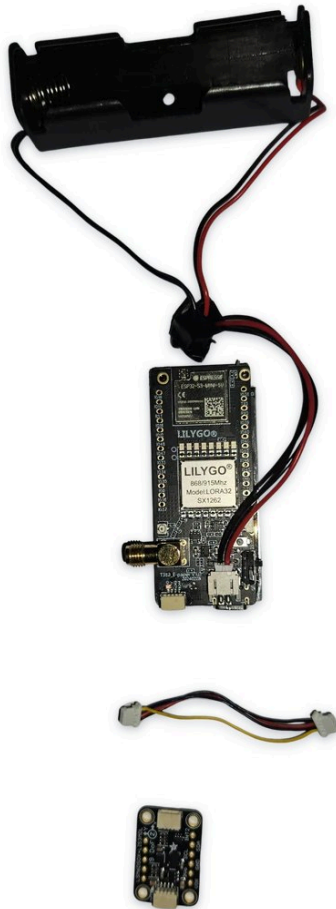
Don't force the plugs. They only fit one way.

5

Connect the IMU

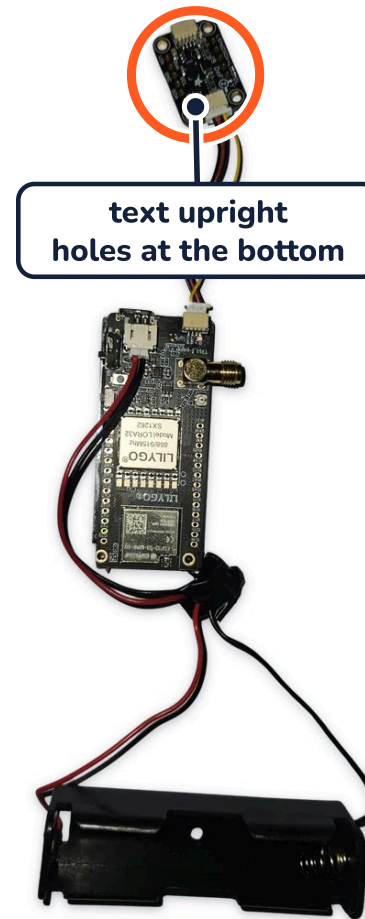


1



Plug a Qwiic cable into the IMU and into the free socket on the LilyGO.

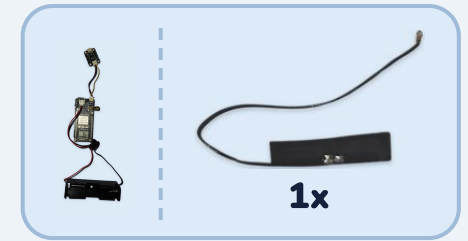
2



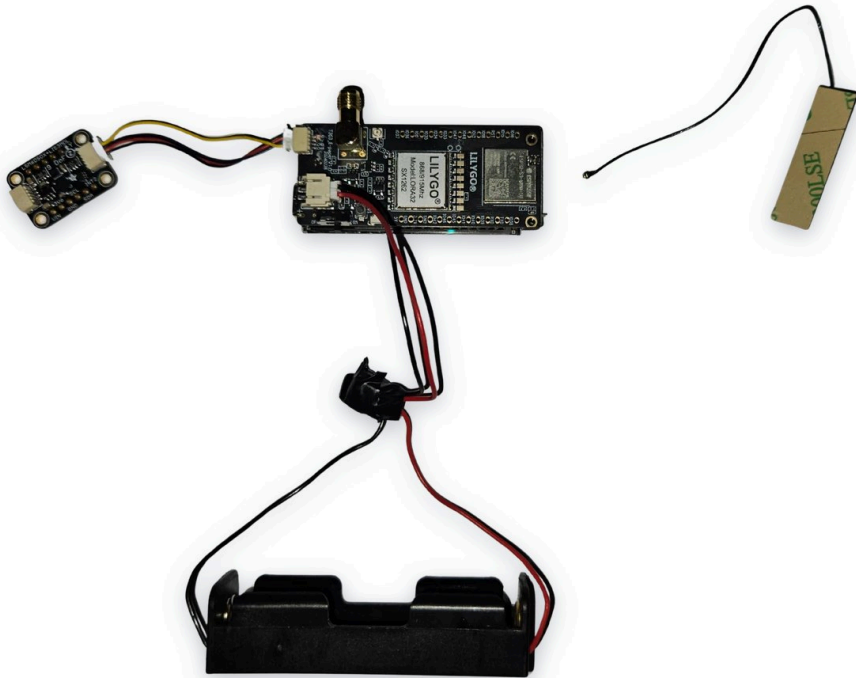
Same rule: text upright, holes at the bottom.

6

Fit the Bluetooth / Wi-Fi antenna

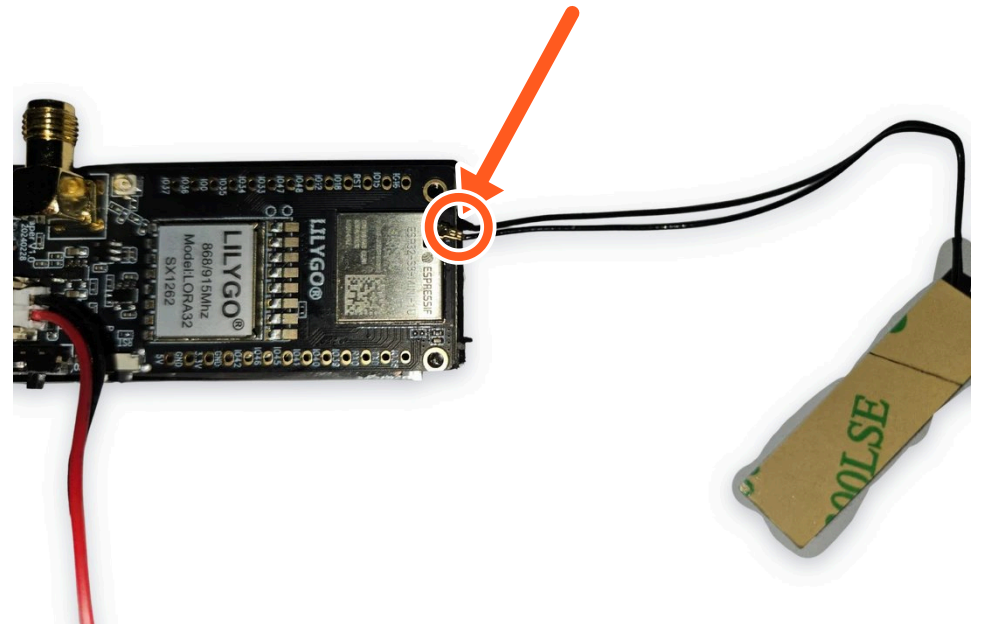


1



Take the small flat antenna.

2



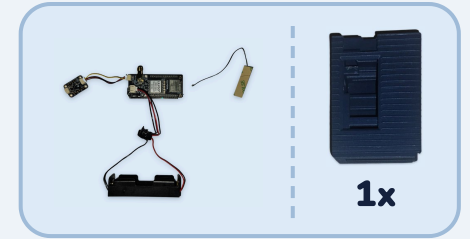
Press its round plug onto the tiny socket by the ESP32 module until it snaps on.



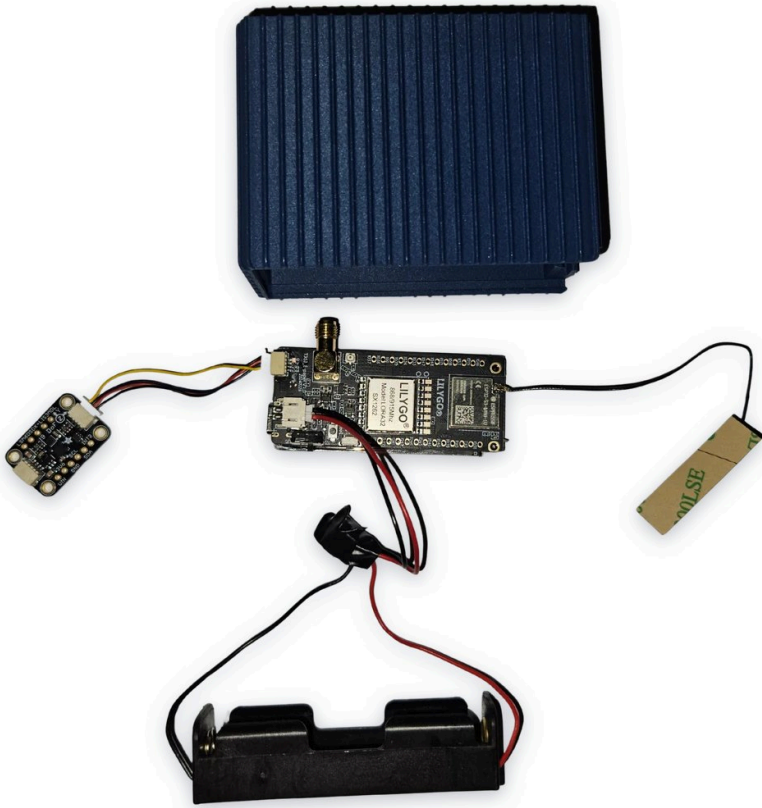
Pull the cable lightly: the plug stays on.

7

Put the board in the case

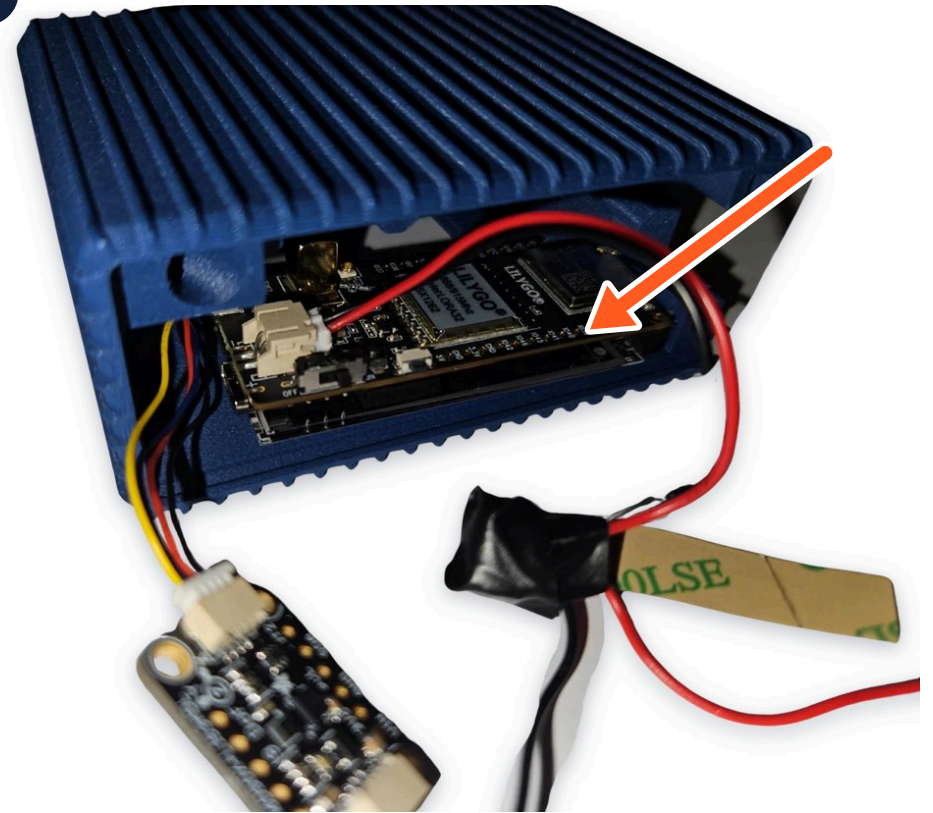


1



Everything you have built, and the case.

2



Screen down, slide the board into its slot. Go gently at first.



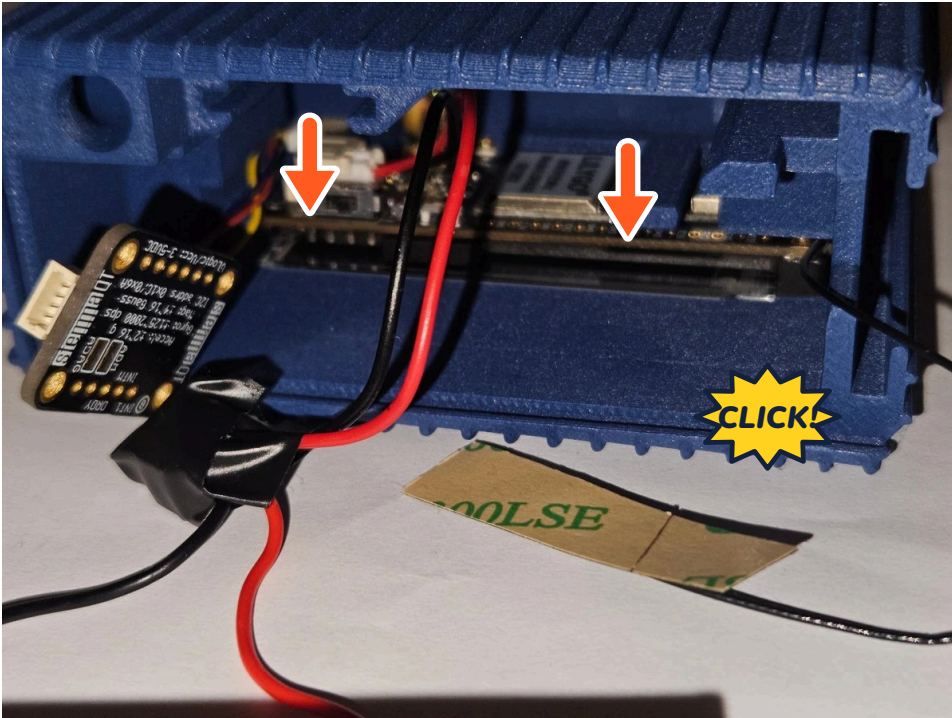
This is the hardest step. Take your time.

7

Put the board in the case

continued

1



Press with two fingers on these two spots until you hear a **click**.

2



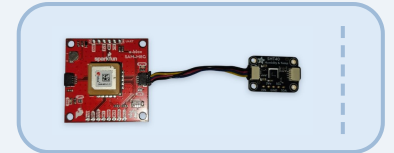
The gold antenna port comes out of the hole on top.



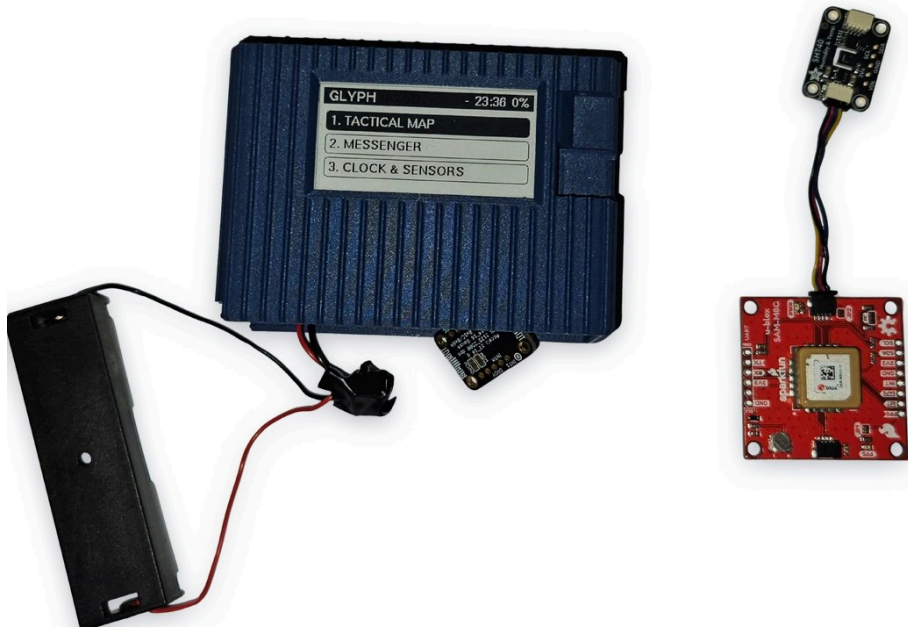
It can take a firm press. Push on the board, never on the screen.

8

Place the GPS and the SHT40

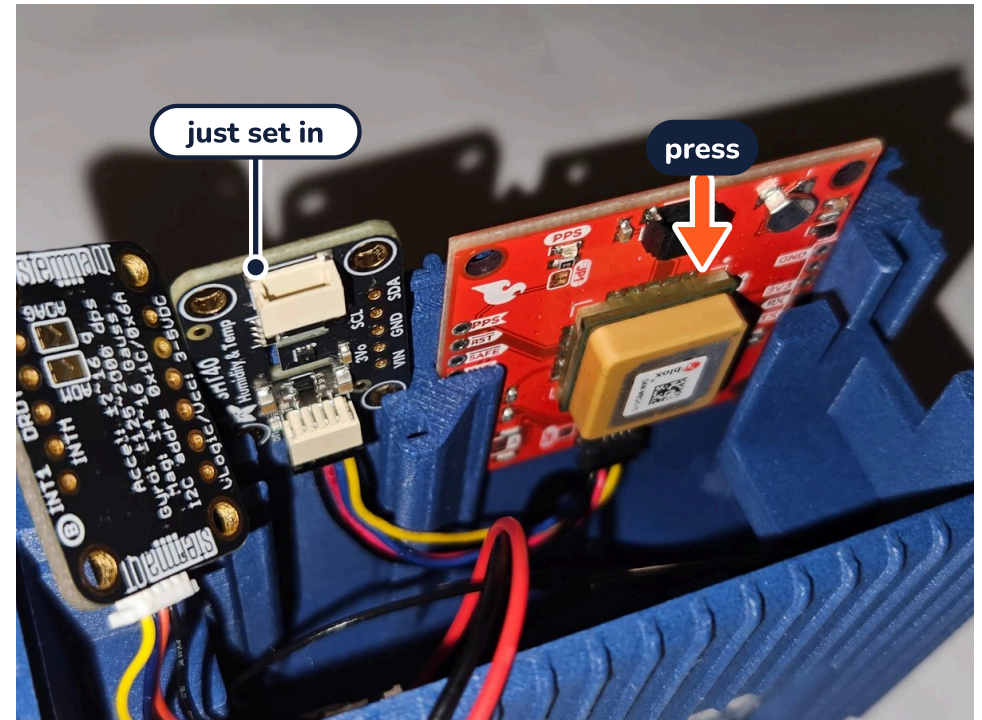


1



Take the GPS and SHT40 you joined in step 4.

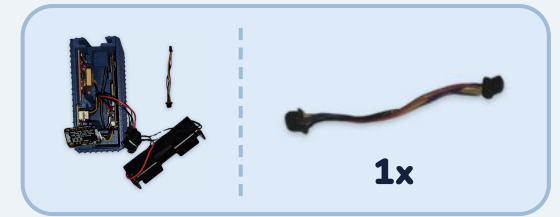
2



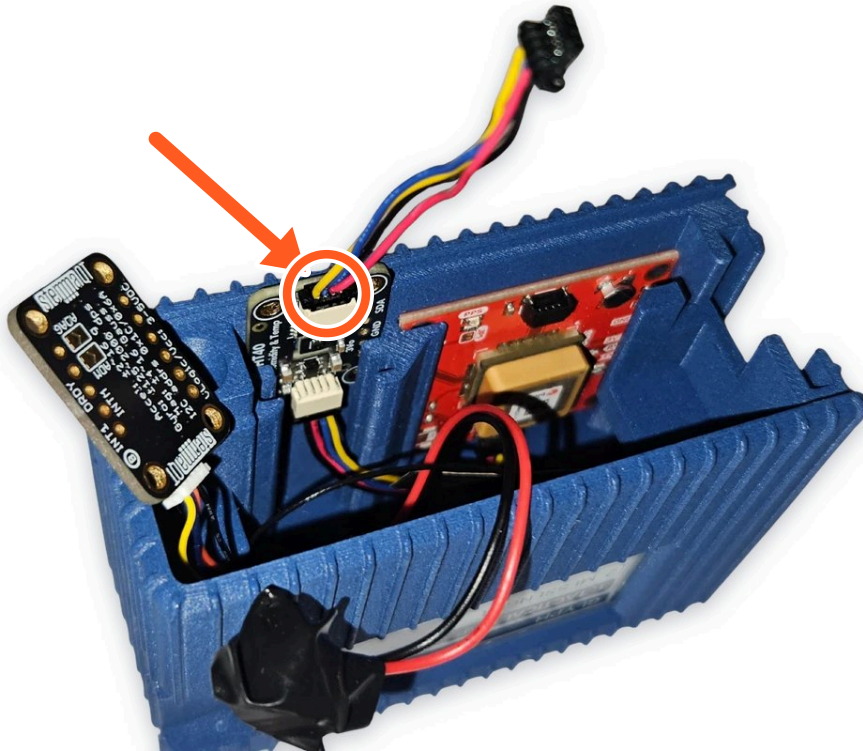
Drop both into their holes, cable at the bottom. Press **only the GPS** all the way down.

9

Connect the SHT40 to the IMU

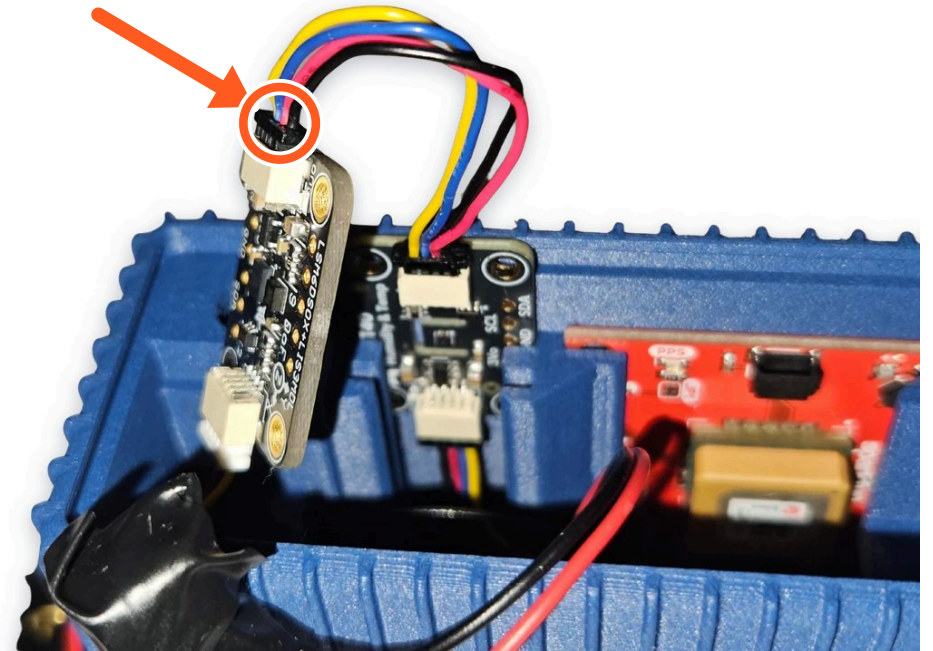


1



Plug a Qwiic cable into the free socket on the SHT40.

2

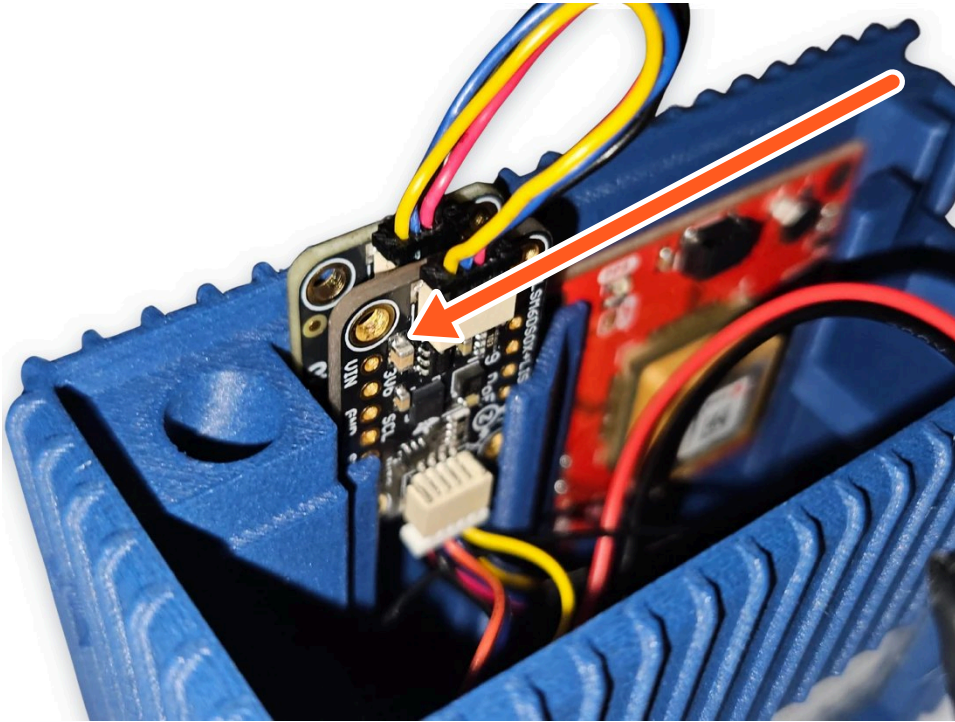


Plug the other end into the IMU.

9

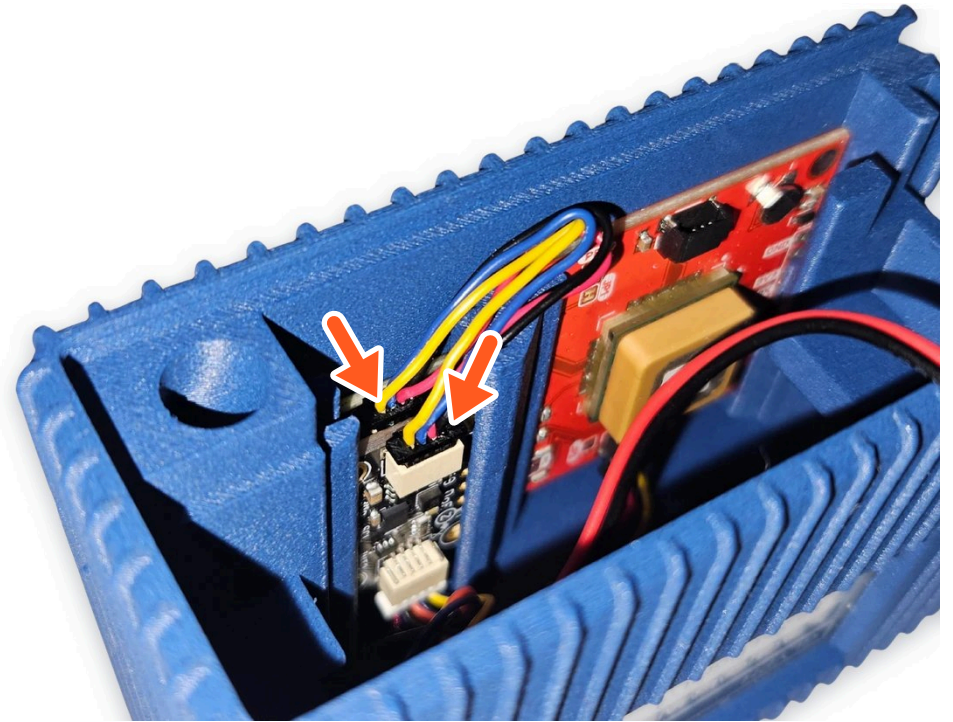
Connect the SHT40 to the IMU

1



Slide the IMU into its slot, next to the SHT40.

2



Press both down a little. They don't need to go all the way.

10 Build the buttons



1x

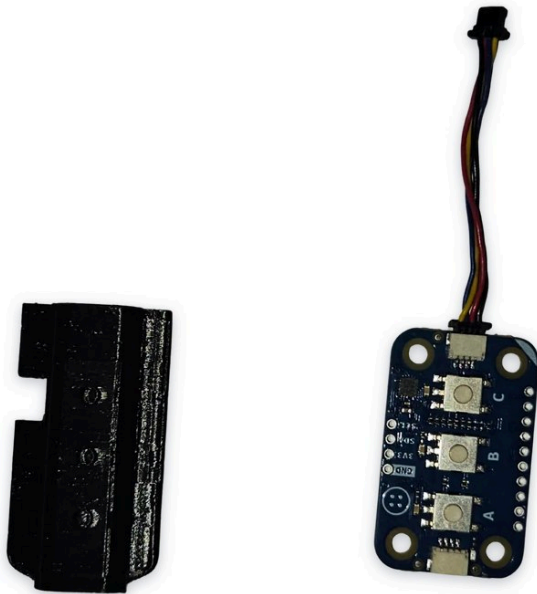


1x



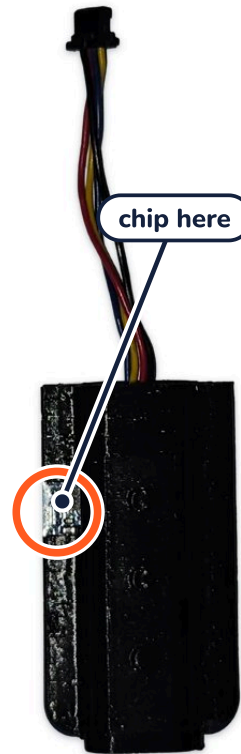
1x

1



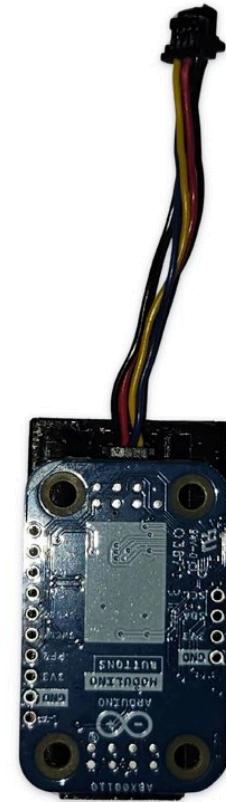
Plug a Qwiic cable into the buttons.
Turn them as shown.

2



Push the TPU cover on. The chip sits at the top right: bend the cable a little so it fits.

3



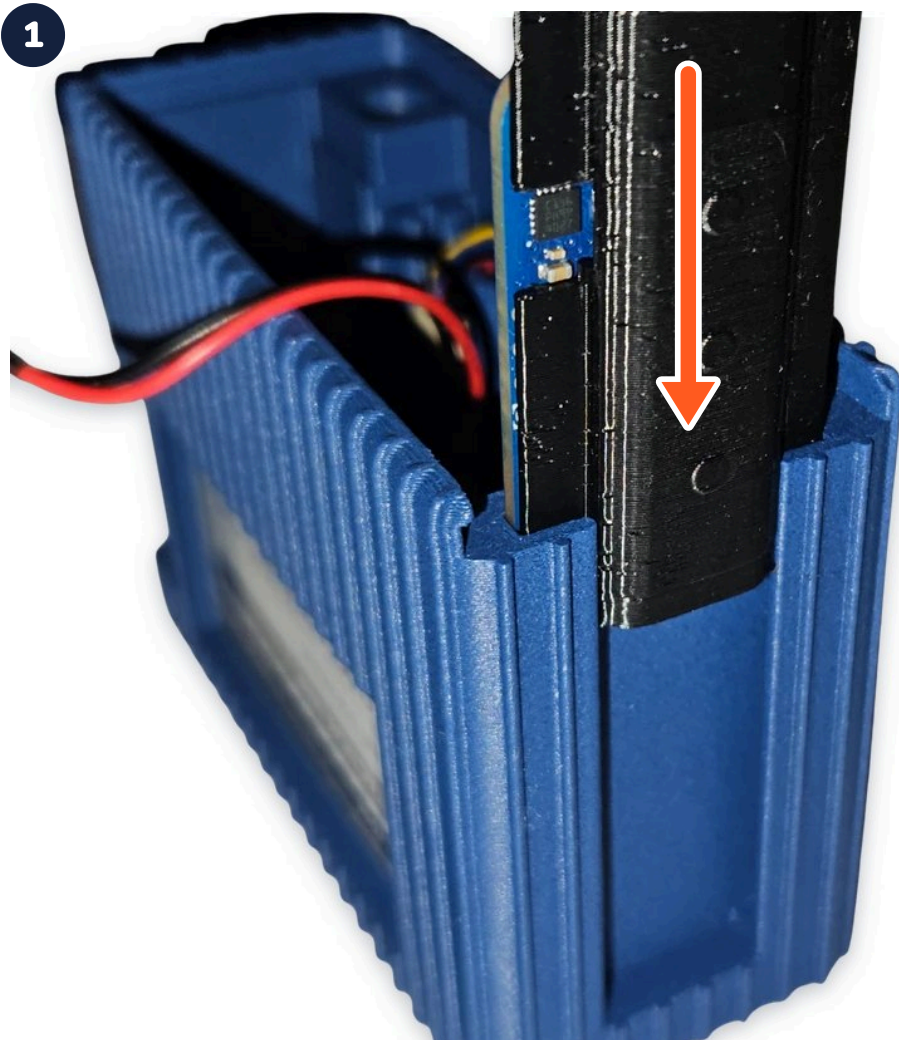
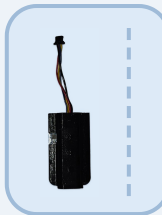
From the back it looks like this.



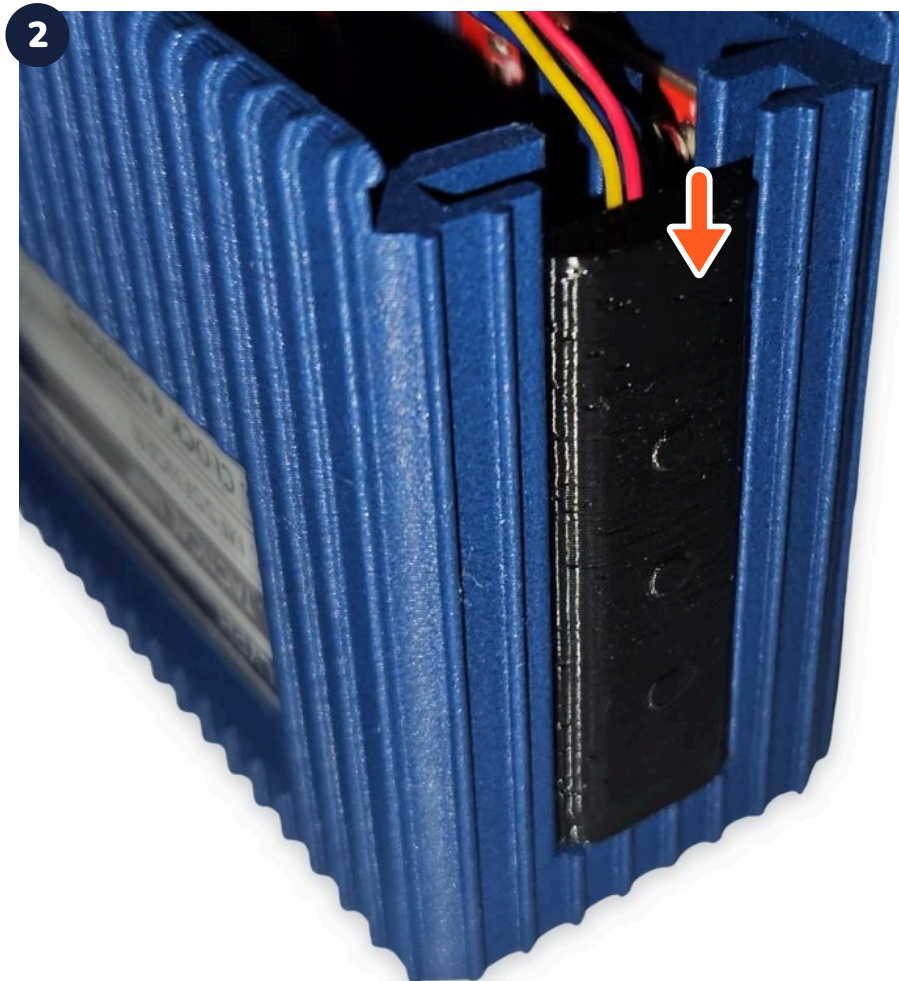
The cover sits flat all round, with no gap.

11

Fit the buttons



Cable up, slide the buttons into the slot in the side of the case.



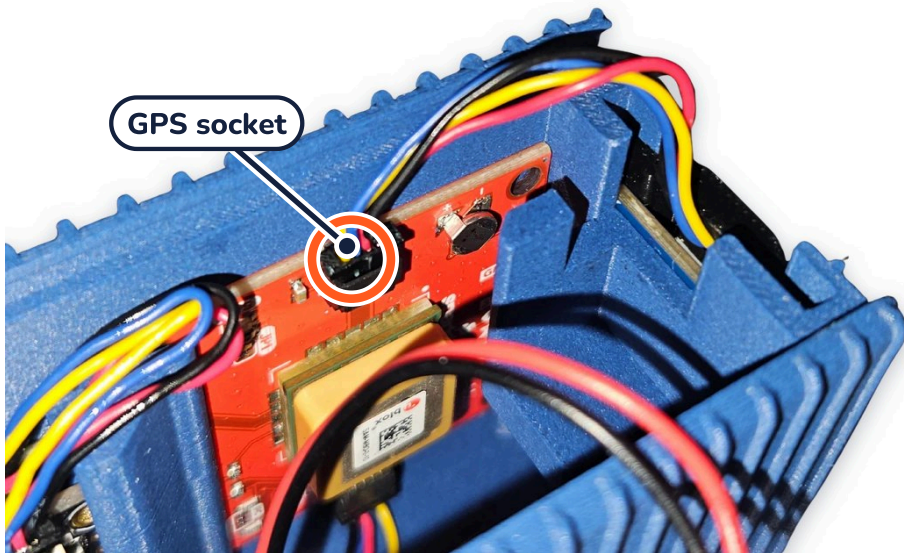
Press down until it stops. It slides in smoothly.

11

Fit the buttons

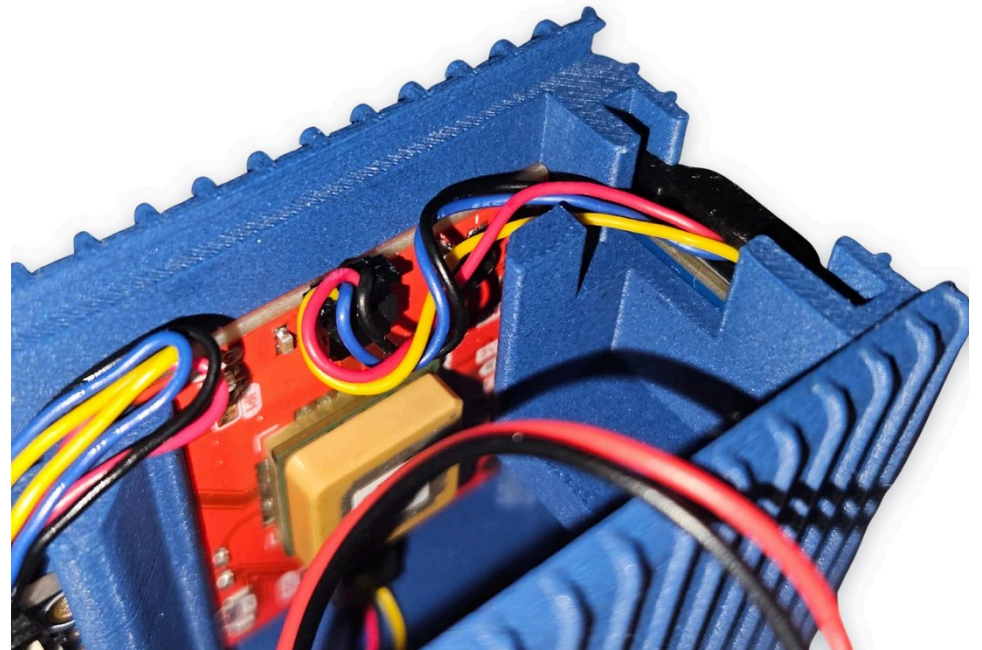
continued

1



Plug the other end of the cable into the GPS.

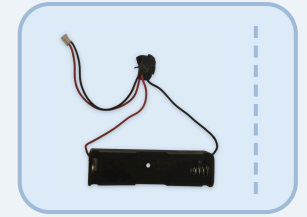
2



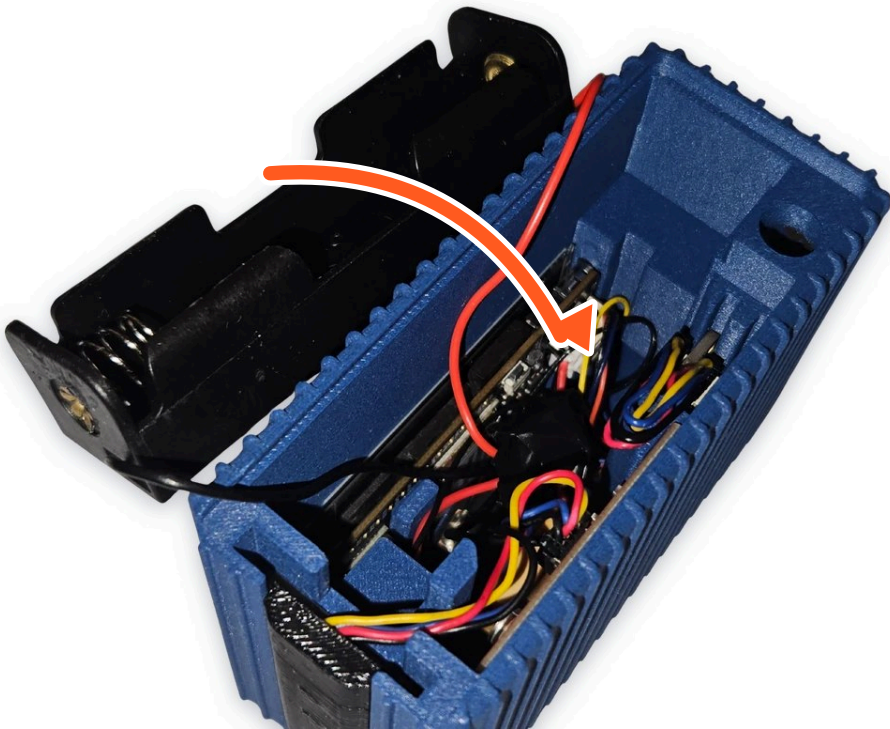
Tuck the cable in like this, clear of the lid.

12

Fit the battery holder

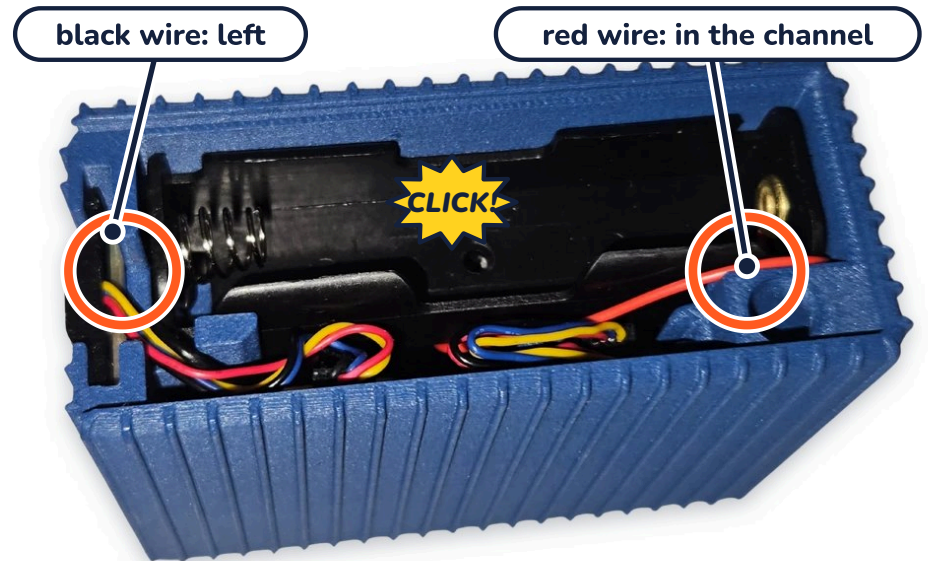


1



Tuck the taped terminal anywhere next to the board, then turn the holder over onto it.

2



Press the holder in until it **clicks**. Black wire on the left, red wire in its channel.

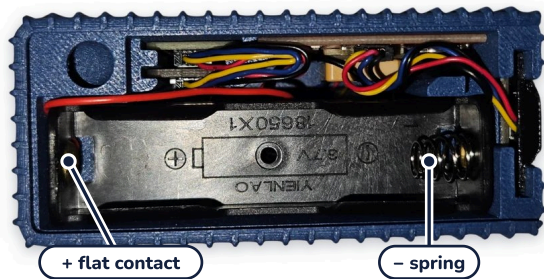
13

Insert the battery



1x

1



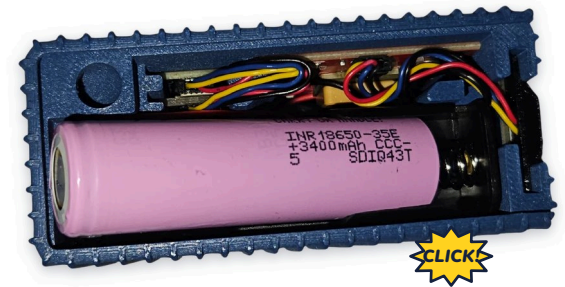
Check the + and – marks in the holder.

2



Flat end of the cell (–) against the spring. Push it in.

3



It **clicks** into place.



Wrong way round can damage GLYPH. Use an undamaged cell with an intact wrap.



Safest: screw on the antenna (step 16) before the cell goes in. A radio that sends without one can damage itself.

14 Close the lid



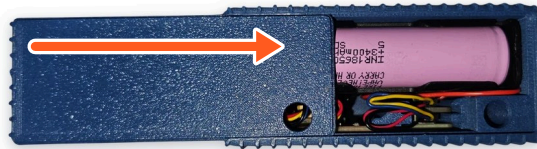
1x

1



Take the lid.

2



Slide it along the rails until it closes.

3



Flush all round.

15 Insert the microSD card



1x

1



Take the microSD card.

2



Slide it into the slot beside the screen.

3



Push until it's all the way in.

16

Screw on the antenna



1x

1



Take the LoRa antenna.

2



Screw it onto the port on top. Finger-tight is enough.

3



Done!



Keep the antenna on whenever GLYPH is switched on: sending without it can damage the radio.



Your GLYPH is built

Next

1. **Install the GLYPH app** on your Android phone, from Google Play or from github.com/Mqr1oo/GLYPH.
2. **Install the firmware:** plug GLYPH into the phone with a USB-C cable, then open the app and follow **Install firmware**. It asks once for the access code that came with your GLYPH files.
3. **Connect** over Bluetooth and give your team a name and a key.

Every GLYPH in your team needs the same team key and the same radio band.

